Raman Sanjay Mankar

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.12
Jawahar Navodaya Vidyalaya , Wardha - 442001	Class XII (CBSE)	2018	85.60%
Jawahar Navodaya Vidyalaya , Wardha - 442001	Class X (CBSE)	2016	88.40%

WORK EXPERIENCE

Micron (Applied AI) | AI/ML Research Intern | May 2026 - July 2026

- Built **SkillOpt**, a self-optimizing **AI skill** for **automated NVMe firmware log error classification**, replacing slow manual log review with an **AI-driven pipeline.**
- Designed an **adversarial optimizer-target model** loop with a validation gate that iteratively refined the skill, **improving classification accuracy** by **18%** on unseen validation logs.
- Contributed the **SkillOpt plugin** to **Micron's Task.ai platform**, packaged for team validation and downstream production testing.
- Delivered **cost-per-run and accuracy benchmarks** across multiple **model configurations**, documenting **architecture trade-offs** to inform business deployment decisions.

TCS (Jaguar Land Rover) | Python Developer -- Project Kubricks | Feb 2024 - Jun 2025

- Developed **GCP Cloud Functions** and event **triggers** automating dataset updates across **JLR data pipelines**.
- Maintained technical documentation in **Confluence** and drove sprint delivery on **Jira** under an **Agile** workflow.

TCS (Jaguar Land Rover) | Python MongoDB Developer -- Project Product 360 | Aug 2023 - Jan 2024

- Optimised **ODS-ODL middleware** data flow using **MongoDB triggers**, increase throughput by **20%** through a document restructuring.
- Guaranteed change-propagation integrity between middleware layers via trigger-based **data validation**.

DTree Labs LLP | Full Stack Developer Intern -- RCC UPSC Portal | March 2023 - May 2023

- Built the **RCC UPSC portal** frontend in **React.js**, **Next.js** and **Mantine UI** with responsive, reusable components.
- Integrated frontend components with backend **REST APIs** for dynamic content delivery.

PROJECTS

GraphRAG-ArXiv | Knowledge Graph + RAG System -- Research Project | Aug 2026 - Sep 2026 | [Github]

- Built a hybrid **Retrieval-Augmented Generation (RAG)** system pairing a **knowledge graph** with **vector search**, routing each query to graph traversal, semantic retrieval, or community summarisation.
- Engineered an **LLM entity-relation extraction** pipeline with **structured outputs**, **multi-stage entity resolution**, and **Leiden community detection**, yielding a **1,130-entity, 1,078-relation** graph with validated citations.
- Cut **token cost 60%** and scored highest on **faithfulness (4.92/5)** against **BM25, dense retrieval, and hybrid RRF** baselines under **LLM-as-a-judge** evaluation, backed by **272 unit tests**.

Bug Tracking System | MCP Server + Claude Agent Plugin | May 2026 - June 2026 | [Github]

- Built a **Claude Agent plugin and MCP server** exposing **11 tools**, **4 skills** and **4 sub-agents** for in-editor **agentic bug triage**.
- Modelled a semiconductor domain in **SQLite** with **8 entities**, **7 chip families** and **10 hardware components**, enforcing a **6-state workflow** with audit history.
- Automated **CVE/CVSS ingestion** from **NVD** and **GitHub Issues** with **explainable AI triage** returning per-field **confidence, evidence and reasoning**.

SecureApp | Secure Full-Stack Banking Web App | Feb 2026 - Mar 2026 | [Github]

- Built a **full-stack banking app** in **PHP, MySQL and Docker** with **15 routes** for **user registration, fund transfers and transaction history**.
- Mitigated **OWASP Top 10** risks (**SQLi, XSS, CSRF**) using **prepared statements**, **256-bit CSRF tokens** and **10+ security headers**.
- Hardened **authentication** with **bcrypt**, a **5/15-attempt lockout** and **session rotation** every **5 minutes**, served over **TLS 1.3**.
- Ensured **race-safe transactions** with **ACID row-level locking** and blocked **malicious uploads** via **image re-encoding**.

Digital Health Coach | Agentic AI WebApp -- Hackathon Project | Feb 2026 - Mar 2026 | [Github]

- Built a **4-agent AI fitness coach** on **FastAPI** and **MongoDB**, serving **7 JWT-authenticated REST endpoints** across **5 indexed collections**.
- Persisted every **agent decision and generated plan** to MongoDB as an **audit trail**, enabling plan-history retrieval and reproducible debugging of non-deterministic LLM outputs.
- Combined **deterministic rule guardrails** -- burnout and inconsistency detection at **< 60% completion rate** and declining mood trends -- with **Mistral LLM** reasoning over **7-day rolling logs** to regenerate adaptive plans.

SKILLS

- **Agentic AI & GenAI:** Agentic AI Systems, LLM Application Development, RAG, GraphRAG, Knowledge Graphs, Vector Databases, Prompt Engineering, Explainable AI, Claude Code, Copilot, Workflow AI Architecture, Agent Team, MCP Server, Plugin, Langchain
- **ML & CV:** PyTorch, scikit-learn, OpenCV, YOLOv8, OCR (PaddleOCR, EasyOCR, Tesseract), CLIP, Vision Transformers, Ensemble Learning, Bagging, Boosting, ML Algorithm, Evaluation, Token Usage, Cost Finding, Huggingface, Roboflow, Kaggle
- **Backend & Data:** FastAPI, Uvicorn, Pydantic, MongoDB, SQLite, ChromaDB, Kuzu, REST APIs, GCP Cloud Functions
- **Frontend & Tools:** React.js, Next.js, Bootstrap, Mantine UI, Git, Jira, Confluence, Vercel, Render, Fromspree, Latex, Google Colab
- **Languages & Soft Skills:** Technical Documentation, Communication, Team Collaboration, Python, JavaScript, TypeScript, SQL, C++

RELEVANT COURSES

Deep Learning | Fundamentals of Machine Learning | Data Mining | Advanced Data Structure and Algorithms | Visual Computing

Advanced Computer Networks | Fraud Analytics | Network Security | Communication Skills | Representation Learning

SCHOLASTIC ACHIEVEMENTS

- **Convolve 2.0 (IIT Guwahati):** Reached the **semi-finals** of the national-level **Generative AI hackathon**.
- **Deep Learning Hackathon:** Secured **3rd place** building a **binary image classifier** for **3D pattern recognition**.

POSITION OF RESPONSIBILITY

- **Handball | National & Zonal Level** -- Represented at zonal and national-level championships.
- **Student Representative Committee | (SRC) Member** -- Represented the student body on academic and campus matters.